8. *Actions for simple systems:* Evaluate the action for the following one-dimensional systems: (a) a free particle (of mass m), (b) a particle (of mass m) that is subjected to a constant force (say, of magnitude α), and (c) a simple harmonic oscillator (of mass m and frequency ω). Assume that these systems are at the points q_1 and q_2 at times t_1 and t_2, respectively.

8) a) $\mathcal{L} = \frac{m\dot{x}^2}{2}$, $J = \int_{t_1}^{t_2} \frac{m\dot{x}^2}{2} dt = \boxed{\frac{mv^2}{2}(t_2 - t_1)}$

b) $m\ddot{x} = \alpha \Rightarrow \dot{x} = \frac{\alpha t}{m} + b \Rightarrow x(t) = \frac{\alpha t^2}{2m} + bt + c$

$V(x) = -\alpha x$

$$J = \int_{t_1}^{t_2} \left(\frac{m\dot{x}^2}{2} + \alpha x\right) dt$$

$$J = \int_{t_1}^{t_2} \left(\frac{m}{2}\left(\frac{\alpha t}{m} + b\right)^2 + \frac{\alpha^2 t^2}{2m} + \alpha bt + c\alpha\right) dt$$

$$J = \int_{t_1}^{t_2} \left[\frac{m}{2}\left(\frac{\alpha^2}{m^2}t^2 + \frac{2\alpha b}{m}t + b^2\right) + \frac{\alpha^2 t^2}{2m} + \alpha bt + c\alpha\right] dt$$

$$J = \int_{t_1}^{t_2} \left(\frac{\alpha^2 t^2}{m} + 2\alpha bt + \frac{mb^2}{2} + c\alpha\right) dt$$

$$J = \frac{\alpha^2 t^3}{3m} + \alpha bt^2 + \left(\frac{mb^2}{2} + c\alpha\right)t \Big|_{t_1}^{t_2}$$

$$\Rightarrow \boxed{J = \frac{\alpha^2}{3m}\left(t_2^3 - t_1^3\right) + \alpha b\left(t_2^2 - t_1^2\right) + \left(\frac{mb^2}{2} + c\alpha\right)(t_2 - t_1)}$$

c) $\mathcal{L} = \frac{m\dot{x}^2}{2} - \frac{1}{2}m\omega^2 x^2$

$x = A\sin(\omega t + \delta)$

$\dot{x} = A\omega\cos(\omega t + \delta)$

$$\mathcal{L} = \frac{mA^2\omega^2}{2}\cos^2(\omega t + \delta) - \frac{1}{2}m\omega^2 A^2 \sin^2(\omega t + \delta)$$

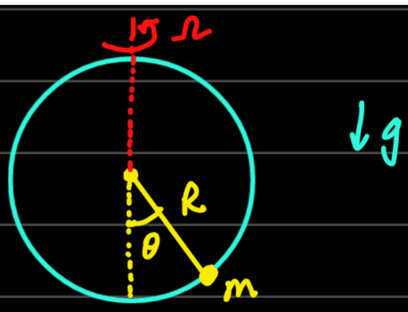
$$\mathcal{L} = \frac{mA^2\omega^2}{2}\cos(2\omega t + 2\delta)$$

$$J = \int_{t_1}^{t_2} \frac{mA^2\omega^2}{2}\cos(2\omega t + 2\delta)\, dt = \frac{mA^2\omega}{4}\sin(2\omega t + 2\delta)\Big|_{t_1}^{t_2}$$

$$= \frac{m\omega A^2}{4}\left(\sin(2\omega t_2 + 2\delta) - \sin(2\omega t_1 + 2\delta)\right)$$

$$\boxed{J = \frac{m\omega A^2}{2} \sin\left(\omega(t_2-t_1)\right) \cos\left(\omega(t_2+t_1) + 2\delta\right)}$$

9. <u>*Bead on a wire:*</u> A bead of mass m slides without friction in a uniform gravitational field on a vertical hoop of radius R. The hoop is constrained to rotate at a fixed angular frequency Ω about its vertical diameter. Let θ denote the position of the bead on the hoop as measured from its lowest point. Write down the Lagrangian for the system and find how the equilibrium values of θ depend on Ω.



$$V_\perp = \Omega R \sin\theta$$

$$\mathcal{L} = \frac{1}{2} m \left(R^2\dot{\theta}^2 + \Omega^2 R^2 \sin^2\theta\right) - mgR(1-\cos\theta)$$

$$\mathcal{L} = \frac{mR^2}{2} \left(\dot{\theta}^2 + \Omega^2 \sin^2\theta\right) + mgR\cos\theta$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{mR^2}{2}\left(2\dot{\theta}\right) = mR^2\dot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{mR^2}{2}\left(2\Omega^2 \sin\theta\cos\theta\right) - mgR\sin\theta$$

$$\Rightarrow \quad mR^2\ddot{\theta} = mR^2\Omega^2 \sin\theta\cos\theta - mgR\sin\theta$$

$$\ddot{\theta} = \Omega^2 \sin\theta\cos\theta - \frac{g}{R}\sin\theta$$

For equilibrium, $\ddot{\theta} = 0$

$$\Rightarrow \quad \sin\theta\left(\Omega^2\cos\theta - \frac{g}{R}\right) = 0$$

$\sin\theta = 0$ or $\cos\theta = \dfrac{g}{\Omega^2 R}$

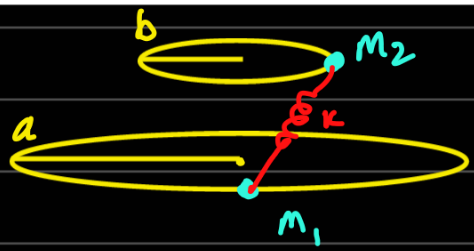
If $\Omega < \sqrt{\frac{g}{R}}$, only $\theta = 0$ and $\theta = \pi$ are equilibrium values

If $\Omega > \sqrt{\frac{g}{R}}$, $\theta = 0$, $\theta = \pi$, $\theta = \pm\cos^{-1}\left(\frac{g}{\Omega^2 R}\right)$ are eq. values.

10. *A coupled system:* Consider two particles of masses m_1 and m_2. Let m_1 be confined to move on a circle of radius a in the $z = 0$ plane, centered at $x = y = 0$. Let m_2 be confined to move on a circle of radius b in the $z = c$ plane, centered at $x = y = 0$. A light and massless spring of spring constant k is attached between the two particles.

(a) Write down the Lagrangian for the system.

(b) Obtain the equations of motion of the system using the method of Lagrange multipliers and give a physical interpretation for each multiplier.



$$\mathcal{L} = \frac{1}{2}m_1 a^2\dot\theta_1^2 + \frac{1}{2}m_2 b^2\dot\theta_2^2 - V$$

$$\bar r_1 = (a\cos\theta_1, a\sin\theta_1, 0)$$
$$\bar r_2 = (b\cos\theta_2, b\sin\theta_2, c)$$
$$\bar r_2 - \bar r_1 = (b\cos\theta_2 - a\cos\theta_1, b\sin\theta_2 - a\sin\theta_1, c)$$

$$V = \frac{1}{2}k|\bar r_2 - \bar r_1|^2 = \frac{1}{2}k\big(b^2\cos^2\theta_2 + a^2\cos^2\theta_1 - 2ab\cos\theta_1\cos\theta_2 + b^2\sin^2\theta_2 + a^2\sin^2\theta_1 - 2ab\sin\theta_1\sin\theta_2 + c^2$$

$$\frac{1}{2}k\left(a^2+b^2+c^2 - 2ab\cos(\theta_1-\theta_2)\right)$$

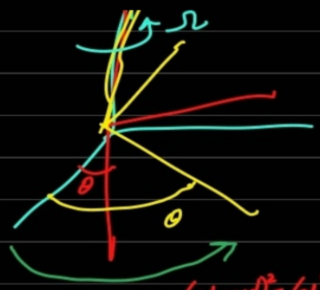
$$\mathcal{L} = \frac{1}{2}m_1 a^2\dot\theta_1^2 + \frac{1}{2}m_2 b^2\dot\theta_2^2 + kab\cos(\theta_1-\theta_2)$$

$$\frac{\partial\mathcal{L}}{\partial\theta_1} = -kab\sin(\theta_1-\theta_2) \qquad \frac{\partial\mathcal{L}}{\partial\dot\theta_1} = m_1 a^2\dot\theta_1$$

$$m_1 a^2\ddot\theta_1 = -kab\sin(\theta_1-\theta_2)$$

$$\boxed{\begin{aligned}\ddot\theta_1 &= -\frac{kb}{m_1 a}\sin(\theta_1-\theta_2)\\ \ddot\theta_2 &= -\frac{ka}{m_2 b}\sin(\theta_2-\theta_1)\end{aligned}}$$

7. *Equations of motion in non-inertial frames:* Obtain the Lagrangian and the equations of motion of a free particle in: (a) a frame that is rotating at a constant angular velocity, say, Ω, with respect to the z-axis, and (b) a frame that is accelerating with a uniform acceleration, say, g, in the direction of positive x-axis.



7) $(x, y, z) \mapsto (x', y', z)$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{pmatrix}$$

$$(a+b+c+d)^2 = (a+b)^2 + (c+d)^2 + 2(a+b)(c+d)$$
$$= a^2+b^2+c^2+d^2 + 2ab + 2cd + 2(ac+ad+bc+bd)$$

$$\begin{pmatrix} \dot{x}' \\ \dot{y}' \end{pmatrix} = \begin{pmatrix} \dot{x}\cos\theta - x\Omega\sin\theta - \dot{y}\sin\theta - y\Omega\cos\theta \\ \dot{x}\sin\theta + x\Omega\cos\theta + \dot{y}\cos\theta - y\Omega\sin\theta \end{pmatrix}$$

$$L = \frac{1}{2} m \left(\dot{x}'^2 + \dot{y}'^2 + \dot{z}^2\right)$$

$$\begin{aligned} = \frac{1}{2} m \Big(& \dot{x}^2\cos^2\theta + x^2\Omega^2\sin^2\theta + \dot{y}^2\sin^2\theta + y^2\Omega^2\cos^2\theta \\ & - 2x\dot{x}\cancel{\Omega}\sin\theta\cos\theta - 2\dot{x}\dot{y}\cancel{\cos\theta}\sin\theta \\ & - 2\dot{x}y\Omega\cos^2\theta \\ & + 2x\dot{y}\Omega\sin^2\theta + 2xy\Omega^2\cancel{\sin\theta}\cos\theta \\ & + 2\dot{y}y\cancel{\Omega}\sin\theta\cos\theta \\ & + \dot{x}^2\sin^2\theta + x^2\Omega^2\cos^2\theta + \dot{y}^2\cos^2\theta + y^2\Omega^2\sin^2\theta \\ & + 2x\dot{x}\cancel{\Omega}\sin\theta\cos\theta + 2\dot{x}\dot{y}\cos\theta\cancel{\sin\theta} \\ & - 2\dot{x}y\Omega\sin^2\theta \\ & + 2x\Omega\dot{y}\cos^2\theta - 2xy\Omega^2\cancel{\sin\theta}\cos\theta - 2y\cancel{\dot{y}}\Omega\sin\theta\cos\theta \\ & + \dot{z}^2 \Big) \end{aligned}$$

$$L = \frac{1}{2} m \left(\dot{x}^2 + x^2\Omega^2 + \dot{y}^2 + y^2\Omega^2 + 2x\dot{y}\Omega - 2\dot{x}y\Omega + \dot{z}^2\right)$$

$$L = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2 + x^2\Omega^2 + y^2\Omega^2 + 2\Omega x\dot{y} - 2\Omega\dot{x}y\right)$$

$$\frac{\partial L}{\partial \dot{x}} = \frac{1}{2} m \left(2\dot{x} - 2\Omega y \right)$$

$$\frac{\partial L}{\partial x} = \frac{1}{2} m \left(2x\Omega^2 + 2\Omega \dot{y} \right)$$

$$\Rightarrow \quad 2\ddot{x} - 2\Omega \dot{y} = 2x\Omega^2 + 2\Omega \dot{y}$$

$$\ddot{x} = \Omega^2 x + 2\Omega \dot{y}$$

$$\frac{\partial L}{\partial \dot{y}} = \frac{1}{2} m \left(2\dot{y} + 2\Omega x \right)$$

$$\frac{\partial L}{\partial y} = \frac{1}{2} m \left(2y\Omega^2 - 2\Omega \dot{x} \right)$$

$$\Rightarrow \quad 2\ddot{y} + 2\Omega \dot{x} = 2\Omega^2 y - 2\Omega \dot{x}$$

$$\Rightarrow \quad \ddot{y} = \Omega^2 y - 2\Omega \dot{x}$$

$$\boxed{\begin{aligned} \ddot{x} &= \Omega^2 x + 2\Omega \dot{y} \\ \ddot{y} &= \Omega^2 y - 2\Omega \dot{x} \\ \ddot{z} &= 0 \end{aligned}}$$

b) $$(x, y, z) \longmapsto \left(x - \frac{gt^2}{2}, y, z \right)$$

$$(\dot{x}, \dot{y}, \dot{z}) \longmapsto \left(\dot{x} - gt, \dot{y}, \dot{z} \right)$$

$$L = \frac{1}{2} m \left(\dot{x}^2 - 2\dot{x}gt + g^2t^2 + \dot{y}^2 + \dot{z}^2 \right)$$

$$\boxed{\ddot{y} = 0 \qquad \ddot{z} = 0 \qquad \ddot{x} = g}$$

$$\frac{\partial L}{\partial \dot{x}} = \frac{1}{2} m \left(2\dot{x} - 2gt \right) = 0$$

$$\frac{1}{2} m \left(2\ddot{x} - 2g \right) = 0 \Rightarrow \quad \ddot{x} = g$$

6. *A different Lagrangian:* A particle of mass m moves in one dimension such that it has the Lagrangian

$$L = \left(\frac{m^2}{12}\right)\dot{x}^4 + m\,\dot{x}^2\,V(x) - V^2(x),$$

where V is a differentiable function of x. Obtain the equation of motion for $x(t)$ and describe the physical nature of the system on the basis of this equation.

$$m^2\dot{x}^2\ddot{x} + 2m\ddot{x}V = -m\dot{x}^2\frac{\partial V}{\partial x} - 2V\frac{\partial V}{\partial x}$$

$$m\ddot{x}\left(m\dot{x}^2 + 2V\right) = -\frac{\partial V}{\partial x}\left(m\dot{x}^2 + 2V\right)$$

$$\Rightarrow \left(m\ddot{x} + \frac{\partial V}{\partial x}\right)\left(\frac{m\dot{x}^2}{2} + V(x)\right) = 0$$

Case 1: $\frac{m\dot{x}^2}{2} + V(x) \neq 0 = E$, $\quad m\ddot{x} = -\frac{\partial V}{\partial x}$

So normal conservative system.

Case 2: $\frac{m\dot{x}^2}{2} + V(x) = 0$, $\quad m\ddot{x} + \frac{\partial V}{\partial x} = c \neq 0$

Diff. wrt t, $\quad m\dot{x}\ddot{x} + \frac{\partial V}{\partial x}\dot{x} = 0 \Rightarrow \dot{x}(c) = 0 \Rightarrow \dot{x} = 0$ identically

$\Rightarrow V(x) = 0$

So free particle at rest.

5. *Parallelepiped inside an ellipsoid:* A rectangular parallelepiped is inscribed in an ellipsoid of semi-axes a, b and c. Using the method of Lagrange multipliers, maximize the volume of the inscribed rectangular parallelepiped. Show that the ratio of the maximum volume to the volume of the ellipsoid is $2/(\sqrt{3}\,\pi)$.

$$f = 8xyz + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$$

$$\frac{\partial f}{\partial x} = 8yz + \frac{2\lambda x}{a^2} = 0 \qquad \frac{\partial f}{\partial y} = 8xz + \frac{2\lambda y}{b^2} = 0 \qquad \frac{\partial f}{\partial z} = 8xy + \frac{2\lambda z}{c^2} = 0$$

$$\lambda = -\frac{4a^2yz}{x} = -\frac{4b^2xz}{y} = -\frac{4c^2xy}{z}$$

$$z = \frac{-4c^2xy}{\lambda}$$

$$y = \frac{-4b^2x}{\lambda}\left(\frac{-4c^2xy}{\lambda}\right) = \frac{16b^2c^2x^2y}{\lambda^2}$$

$$\Rightarrow \quad x^2 = \frac{\lambda^2}{16b^2c^2}$$

$$z = \frac{-4c^2}{\cancel{\lambda}} \cdot \frac{\cancel{\lambda}}{\cancel{4}bc}\, y$$

$$x = \frac{\lambda}{4bc}$$

$$z = -\frac{c}{b}y$$

$$x = \frac{-4a^2y\cancel{z}}{\lambda}$$

$$\frac{\lambda}{4bc} = \frac{4a^2}{\lambda}\, y\, \frac{c}{b}y$$

$$\frac{3\lambda^2}{16b^2c^2a^2} = 1$$

$$\frac{\lambda}{4bc} = \frac{4a^2c}{\lambda b}\, y^2$$

$$\lambda^2 = \frac{16b^2c^2a^2}{3}$$

$$\frac{\lambda^2}{16a^2c^2} = y^2$$

$$\lambda = \frac{4abc}{\sqrt{3}}$$

$$y = \frac{\lambda}{4ac}$$

$$V = 8 \cdot \frac{\lambda}{4bc} \cdot \frac{\lambda}{4ac} \cdot \frac{\lambda}{4ab}$$

$$z = -\frac{\lambda}{4ab}$$

$$= \frac{1}{8}\,\frac{\lambda^3}{a^2b^2c^2} = \frac{1}{8a^2b^2c^2}\,\frac{64a^3b^3c^3}{3\sqrt{3}} = \boxed{\frac{8abc}{3\sqrt{3}}}$$

$$V_e = \frac{4}{3}\pi abc \qquad \therefore \frac{V_1}{V_e} = \frac{\overset{2}{\cancel{8}}\,\cancel{abc}\cdot 3}{3\sqrt{3}\,\cancel{4}\,\pi\,\cancel{abc}} = \boxed{\frac{2}{\pi\sqrt{3}}}$$

4. *Fermat's principle in inhomogeneous media:* Consider a light ray propagating between two points, say, (x_1, y_1) and (x_2, y_2), in a two-dimensional inhomogeneous medium whose refractive index is described by the function $n(y, x)$. The velocity of light at any given point in the medium is $c/n(y, x)$ and, hence, according to the Fermat's principle, the light ray will follow a path between the two points such that the integral

$$I = \int_{(x_1,y_1)}^{(x_2,y_2)} \mathrm{d}l \, n(y, x)$$

is a minimum. (Note that $\mathrm{d}l$ denotes the infinitesimal arc length.) For $x_1 = -x_2 = -1$ and $y_1 = y_2 = 1$, find the path $y(x)$ of the light ray in media for which $n = e^y$ and $n = a\,(y - y_0)$, where $y_0 > 0$.

$$I = \int \sqrt{1+y_x^2}\; n(y,x)\, dx$$

1 So $f(x, y, y_x) = \sqrt{1+y_x^2}\; n(y,x) = \sqrt{1+y_x^2}\; e^y$

Using alternate form of EE,

$$\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y_x \frac{\partial f}{\partial y_x}\right) = 0$$

$$\Rightarrow \quad 0 - \frac{d}{dx}\left(\sqrt{1+y_x^2}\; e^y - \frac{y_x^2 e^y}{\sqrt{1+y_x^2}}\right) = 0$$

$$\Rightarrow \quad \frac{e^y}{\sqrt{1+y_x^2}} = \text{constant} = \alpha$$

$$e^{2y} = \alpha^2 + \alpha^2 y_x^2$$

$$\sqrt{\frac{e^{2y}-\alpha^2}{\alpha^2}} = \frac{dy}{dx}$$

$$\Rightarrow \quad \frac{dy}{\sqrt{e^{2y}-\alpha^2}} = \frac{dx}{\alpha}$$

$$e^{2y} - \alpha^2 = u^2 \quad \Rightarrow \quad e^{2y}\, dy = u\, du$$

$$\frac{e^{2y}\, dy}{e^{2y}\sqrt{e^{2y}-\alpha^2}} = \frac{dx}{\alpha}$$

$$\Rightarrow \frac{\cancel{u}\, du}{(u^2+\alpha^2)\cancel{u}} = \frac{dx}{\alpha}$$

$$\Rightarrow \int \frac{du}{u^2+\alpha^2} = \int \frac{dx}{\alpha}$$

$$u = \alpha \tan\left(\frac{x}{\alpha}+c\right)$$

But $u = \sqrt{e^{2y}-\alpha^2}$

$$\Rightarrow e^{2y} = \alpha^2 \sec^2\left(\frac{x}{\alpha}+c\right)$$

$$e^y = \alpha \sec\left(\frac{x}{\alpha}+c\right)$$

$$\text{or} \quad y = \log\left(\sec\left(\frac{x}{\alpha}+c\right)\right) + \log\alpha$$

Given this passes through $(-1,1)$ and $(1,1)$

$$1 = \log\left(\sec\left(\frac{-1}{\alpha}+c\right)\right) + \log\alpha \qquad 1 = \log\left(\sec\left(\frac{1}{\alpha}+c\right)\right) + \log\alpha$$

$c = 2k\pi$ works

$$\Rightarrow 1 = \log\left(\alpha\sec\left(\frac{1}{\alpha}\right)\right) \Rightarrow \alpha\sec\left(\frac{1}{\alpha}\right) = e$$

$\alpha \approx 0.782$ works

$$\therefore \boxed{y = \log\left(0.782\sec\left(\frac{x}{0.782}+2k\pi\right)\right)}$$

② $n(y,x) = a(y-y_0)$

$$f(x,y,y_x) = a(y-y_0)\sqrt{1+y_x^2}$$

Again, $$\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y_x \frac{\partial f}{\partial y_x}\right) = 0$$

$$\Rightarrow \quad f - y_x \frac{\partial f}{\partial y_x} = \alpha$$

$$a(y-y_0)\sqrt{1+y_x^2} - \frac{a(y-y_0)y_x^2}{\sqrt{1+y_x^2}} = \alpha$$

$$a(y-y_0) = \alpha\sqrt{1+y_x^2}$$

$$\sqrt{\frac{a^2}{\alpha^2}(y-y_0)^2 - 1} = \frac{dy}{dx}$$

$$\frac{dx}{\alpha} = \frac{dy}{\sqrt{a^2(y-y_0)^2 - \alpha^2}}$$

$$a(y-y_0) = u$$

$$\Rightarrow \quad a\,dy = du$$

$$\frac{a}{\alpha}dx = \frac{du}{\sqrt{u^2-\alpha^2}}$$

$u = \alpha\cosh s$

$du = (\alpha\sinh s)\,ds \Rightarrow \quad \frac{a}{\alpha}dx = \frac{\alpha\sinh s}{\alpha\sinh s}ds$

$$\Rightarrow \int \frac{a}{\alpha}dx = \int ds$$

$c + \frac{ax}{\alpha} = s$ But $s = \cosh^{-1}\left(\frac{u}{\alpha}\right)$

$$\Rightarrow \quad u = \alpha\cosh\left(\frac{ax}{\alpha} + c\right)$$

and $u = a(y-y_0)$

$$y = y_0 + \frac{\alpha}{a}\cosh\left(\frac{ax}{\alpha}+c\right)$$

Let $\frac{\alpha}{a} = \gamma \Rightarrow y = y_0 + \gamma\cosh\left(\frac{x}{\gamma}+c\right)$

This passes through $(-1,1)$ and $(1,1)$

$$1 = y_0 + \gamma\cosh\left(-\frac{1}{\gamma}+c\right) \qquad 1 = y_0 + \gamma\cosh\left(\frac{1}{\gamma}+c\right)$$

$$\Rightarrow -\frac{1}{\gamma}+c = \frac{1}{\gamma}+c \quad \text{or} \quad -\frac{1}{\gamma}+c = -\left(\frac{1}{\gamma}+c\right)$$

Not valid $\qquad \Rightarrow \underline{c=0}$

So $1 = y_0 + \gamma\cosh\left(\frac{1}{\gamma}\right) \Rightarrow \gamma\cosh\left(\frac{1}{\gamma}\right) = 1-y_0$

$$\therefore \boxed{y = y_0 + \gamma\cosh\left(\frac{x}{\gamma}\right) \text{ where } \gamma\cosh\left(\frac{1}{\gamma}\right) = 1-y_0}$$

1. *Back to the brachistochrone problem:* Recall that, the solution to the brachistochrone problem consisted of extremizing integrals of the type

$$I = \int_{x_1}^{x_2} \frac{\mathrm{d}x}{\sqrt{2\,g\,y}} \left(1 + y_x^2\right)^{1/2} = \int_{y_1}^{y_2} \frac{\mathrm{d}y}{\sqrt{2\,g\,y}} \left(1 + x_y^2\right)^{1/2},$$

where in the first integral y and x assume their usual roles, i.e. y is the dependent variable and x the independent variable, whereas their roles are reversed in the second integral (with $x_y = \mathrm{d}x/\mathrm{d}y$.) Also, we had obtained the solution to the brachistochrone problem by solving the Euler's equation corresponding to the second integral. Using the first integral and the second form of the Euler's equation in the last problem, obtain the solution to the brachistochrone problem.

$$f(x, y, y_x) = \frac{\sqrt{1 + y_x^2}}{\sqrt{2gy}}$$

Using alternate EE,

$$f - y_x \frac{\partial f}{\partial y_x} = c$$

$$\frac{\sqrt{1 + y_x^2}}{\sqrt{2gy}} - \frac{y_x^2}{\sqrt{2gy}\sqrt{1+y_x^2}} = c$$

$$1 = c\sqrt{2gy}\sqrt{1+y_x^2}$$

$$y_x^2 = \frac{1 - 2c^2 gy}{2c^2 gy}$$

$$\frac{dy}{dx} = \sqrt{\frac{1-2c^2 gy}{2c^2 gy}}$$

But this was done earlier as well.